

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

ANSWERS

CLASS

INDEX NUMBER

BIOLOGY

9648/03

Paper 3 Applications Paper and Planning Question

11 September 2017

2 hours

Additional Materials: Answer Paper
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	16
2	11
3	13
4	12
5	20
Total	72

This document consists of **14** printed pages and **2** blank pages.



Answer **all** questions.

- 1 Erythropoietin is a glycoprotein hormone which plays an important role in erythropoiesis (production of erythrocytes) from haematopoietic stem cells. It is a cytokine that binds erythrocyte precursors in the bone marrow

- (a) Describe how red blood cells are formed from haematopoietic stem cells.

1. (multipotent) HSC undergo asymmetrical div → 1 cell remains as SC, the o/r becomes progenitor cell

myeloid progenitor cell divide & dy/dx into erythroid lineage

2. cell lose nucleus

↑ expression of globin genes to ↑ Hb

[2]

The erythropoietin gene can be cloned into bacteria host cell for production of the protein using genetic engineering. Erythropoietin is administered to anaemic patients who are unable to produce sufficient erythrocytes. Fig. 1.1 shows the human erythropoietin gene ligated with *Bam*HI and *Hind*III linkers.

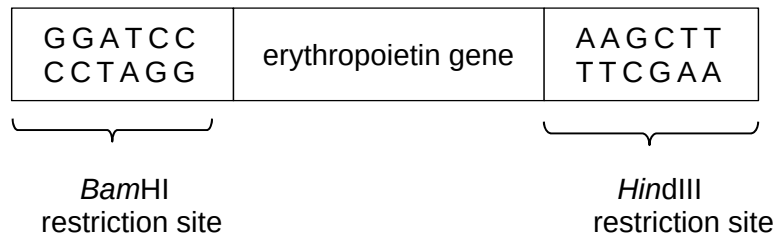


Fig. 1.1

- (b) *Bam*HI cuts between guanine nucleotides while *Hind*III cuts between adenine and guanine nucleotides.

- (i) Illustrate the resultant restriction fragment from Fig. 1.1 after restriction digestion by *Bam*HI and *Hind*III.



[2]

- (ii) Suggest **two** reasons why the erythropoietin gene is cut using two different restriction enzymes.

1. ensure that gene is inserted in the correct orientation

downstream of promoter → correct prot produced

2. prevents gene from circularizing

as the 2 sticky ends are not complementary

[2]

- (iii) Explain why the use of restriction enzymes like *Bam*HI has a higher chance of producing recombinant DNA with plasmid compared to *Sma*I which cuts symmetrically at its restriction site.

1. ***Bam*HI produces sticky ends → anneal with plasmid cut with the same RE via H bonds**

gene & plasmid are held in correct orientation for longer periods for ligase to ligate the DNA

2. ***Sma*I produces blunt ends**

gene & plasmid may detach before ligase can bind ® gene & plasmid cannot ligate

[2]

- (c) Both *Hind*III and *Bam*HI are restriction enzymes commonly found in bacteria.

- (i) Describe the natural function of restriction enzymes.

1. **protect bact from infection by phage**

by recog & binding to specific base seq called RE sites

2. **to hydrolyse phosphodiester bonds btw adj nucleotides**

thus cleaving / digesting foreign DNA

[2]

- (ii) Explain why the bacterium's own DNA is not digested by its own restriction enzymes.

1. **methylates specific bases (e.g. C) in the recognition seq**

thus Δ ing the 3D config of the restriction site

2. **RE site no longer complementary to RE active site**

unable to form ES cplx

[2]

- (d) (i) The human erythropoietin gene was isolated from the genomic library and introduced into a bacterium host cell. The resultant erythropoietin protein produced was non-functional and found to be longer than normal.

Explain why the erythropoietin protein produced by bacteria was longer than normal.

1. **presence of (non-coding seq) introns in erythropoietin gene**

bact $\otimes$ splice off due to lack of post-transcriptional modification

2. **ribo reads introns as codons**

translates them to give extra aas (in the pri prot struct)

[2]

- (ii) Another copy of the erythropoietin gene was isolated from cDNA library. Subsequent introduction of this erythropoietin gene into *E. coli* produced erythropoietin of the correct length but is still non-functional.

Suggest why this is so.

1. erythropoietin prot is a glycoprot

where carboh chains are added to prot in ER / GA by glycosylation

2. however, bact lacks mem-bound organelles like GA to perform post-translational modification

thus prot produced w/o carboh chain is non-functional

[2]

[Total: 16]

2 Read the following passage.

Can't go for liver transplant? There's hope still.

(Excerpt from The Straits Times, 12 Jul 2017)

A transplant can cure end-stage liver cirrhosis - hardening of the liver - but not all patients have this option. For those who are not eligible for a transplant, an alternative may be in sight. 1

A clinical trial was launched yesterday to explore the use of stem cells to reverse liver cirrhosis. In the study conducted by a multi-centre team led by the National University Hospital (NUH), doctors aim to determine if stem-cell therapy can improve liver function. Stem cells will be taken from a patient's own bone marrow and will be isolated and injected directly into the patient's liver to initiate the repair. 5

The \$2.6 million Phase III trial will use biopsy and clinical measurements of liver function to test the efficacy, effectiveness and safety of the stem-cell treatment. The study is funded by the National Medical Research Council, and a total of 46 patients will be recruited. 10

Liver cirrhosis is caused by diseases such as chronic hepatitis B and non-alcoholic fatty liver disease. A liver transplant provides a definitive cure to end-stage cirrhosis. However, in Singapore, less than 5% of end-stage liver cirrhosis patients receive a liver transplant. 15

The number of people on the waiting list for a liver transplant has been increasing over the years, according to statistics from the Ministry of Health. Last year, there were 57 on the waiting list, up from 9 in 2007. There are around 50 waiting for a liver transplant this year. 17

While similar therapy treatments have been conducted overseas in countries such as Egypt and India, they have not been fully evaluated for efficacy. 20

Associate Professor Dan Yock Young, a senior consultant in the division of gastroenterology and hepatology at NUH, said: "We are conducting the study in a systematic and scientific manner in order to get definitive evidence of the effects of the treatment." 25

He said stem-cell therapy is not a substitute for a liver transplant but provides an option for those who are not eligible for one.

- (a) With reference to the features of stem cells, explain how stem-cell therapy can improve liver function in patients (line 6).

1. is multipotent

capable of dy/dx into diff specialised cells

2. unlimited replicative capacity (due to active telomerase)

capable of mitotic division to provide new liver cells for the rest of the patient's lifetime

3. display plasticity

able to undergo transdifferentiation to produce cell type other than blood cells

[3]

- (b) Explain why stem cells are taken

- (i) the bone marrow (line 7).

1. accessibility

ease to obtain in less obstructive manner to patient

2. relatively abundant SC

c.f. to other organs

[2]

- (ii) from the patients themselves (line 7).

1. to prevent immune cells / WBCs from recognising transplanted stem cells

as foreign / antigen ® tissue rejection as no tissue is transplanted

[1]

- (c) Suggest why less than 5% of end-stage liver cirrhosis patients receive a liver transplant (line 15).

1. lack of willing live donors

2. organs from diseased individuals cannot be used for fear of infection to the recipient

3. histologically / genetically incompatibility btw donor and recipient leading to organ rejection

(any 1 MP)

[1]

- (d) Suggest why stem-cell therapy is not a substitute for a liver transplant (line 25).

1. lack of definitive evidence of the effects of the treatment

may lead to undesirable long term side effects

2. efficacy / effect of stem-cell therapy may be transient

(any 1 MP)

[1]

Haemochromatosis is an autosomal recessive disease that leads to excessive iron absorption from diet leading to abnormal deposition in liver and other organs. The iron built up can damage liver cells leading to liver cirrhosis.

The HFE gene at locus 6p22.2 undergoes C282Y mutation that results in 2 instead of 1 *Rsa*I recognition site.

- (e) A DNA fragment was extracted from a HFE carrier, subjected to polymerase chain reaction (PCR), followed by *Rsa*I restriction digestion and separated using gel electrophoresis. The resultant gel is shown in Fig. 2.1.

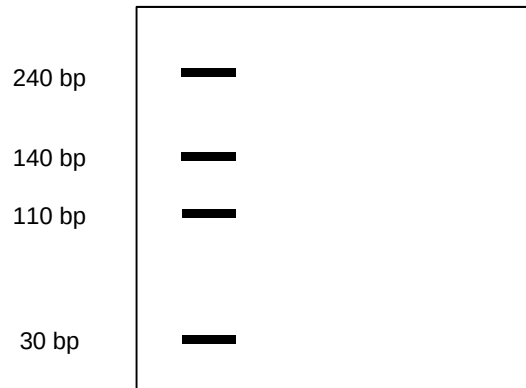
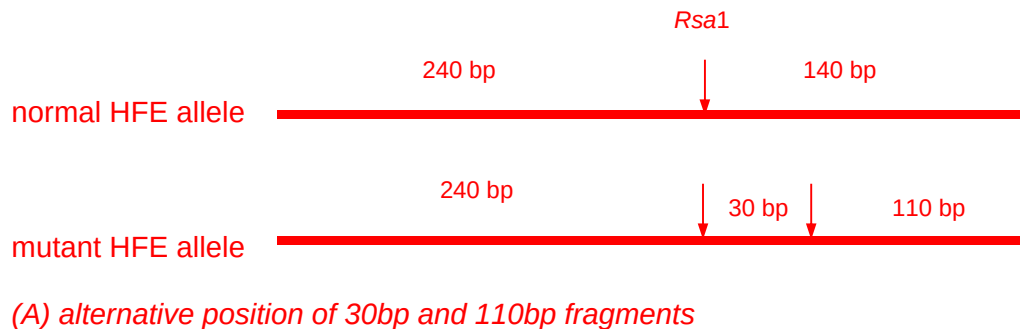


Fig. 2.1

- (i) Illustrate the restriction map for both the normal and mutant HFE alleles.



[2]

- (ii) Calculate the number of PCR cycles needed to obtain 1 million copies of the HFE allele from one copy.

$$2^n = 1\,000\,000$$

$$n \lg 2 = \lg 1\,000\,000$$

$$n = 19.9$$

20 cycles are needed

[1]

[Total: 11]

3 Cultivated rice, *Oryza sativa*, is often grown in fields flooded with water.

Some varieties of cultivated rice are able to grow long internodes (a length of stem between leaves) when they are submerged in water, keeping the leaves and flowers above water level. These varieties are known as deepwater rice. The snorkel genes SK1 and SK2, thought to be responsible for this response, were identified in a variety of deepwater rice, C9285. A non-deepwater variety, T65, did not have these genes.

The expression of SK1 and SK2 genes result in increased production of gibberellin, GA. GA is a plant growth regulator which promotes cell growth and elongation. Fig. 3.1 shows the effect of submergence on GA production in C9285 and in T65.

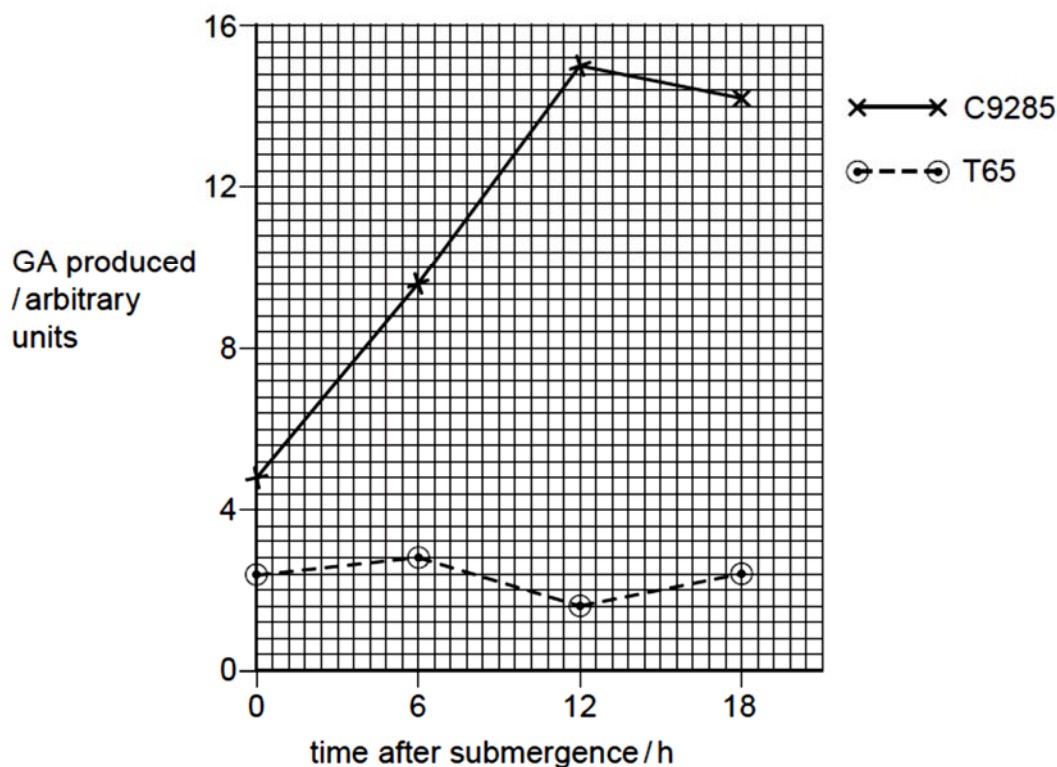


Fig. 3.1

(a) Compare the effect of submergence on GA production in C2985 and T65.

1. **[compare] GA production is higher in C2985 than T65**

for all length of submergence from 0 – 18 hrs

2. **[QV] ↑ GA production in C9285 from 4.8 to max. of 15 units**

while GA levels in T65 remain approx. constant btw 1.6 to 2.8 units

[2]

The effect of duration of submergence of rice plants was also investigated. Fig. 3.2 shows the result on internode elongation.

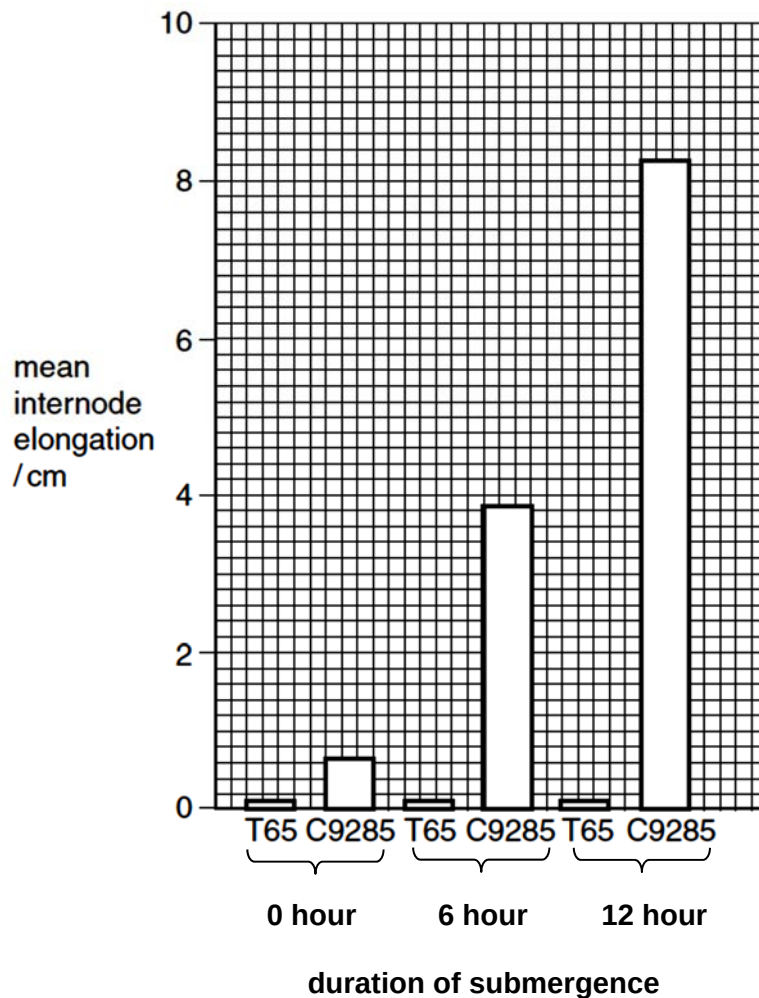


Fig. 3.2

- (b) With reference to Fig. 3.2 and the information given, explain the effect of submergence on internode elongation in T65 and C9285.
- [describe] ↑ length of submergence from 0 – 12 hrs ↑ mean internode elongation in C9285 from 0.7 cm to 8.3 cm**
but has no effect on T65 which remains constant at 0.1 cm
 - [explain] ↑ length of submergence ➔ ↑ expression of SK1 and SK2 in C9285**
leading to ↑ GA production thus promoting cell growth & elongation ➔ ↑ internode elongation [2]
- (c) Deepwater rice is the main food crop in many parts of the world that undergo flooding in the rainy season. However, many varieties of deepwater rice have lower yields than non-deepwater varieties. Deepwater rice variety with high yield can be produced using genetic engineering technology. However, farmers attempted to produce such variety using artificial selection instead.

- (i) Describe how cells from high yield rice plant can be genetically modified to have deepwater characteristics.

1. isolated SK1 and SK2 genes from deepwater variety using RE

ligated into Ti plasmid (cut w same RE) with antibiotic resistant gene (as selectable marker)

2. transform recombinant plasmid into Agrobacterium via heat shock

select for transformed, recombinant bact cells (using replica plating)

3. infect plant cells from high yield variety with recombinant Agrobact

select for infected / transformed plant cells which grows successfully in antibiotic medium

[3]

- (ii) Describe how genetically modified rice plant cells in (c)(i) can be grown into rice plants.

1. via tissue culture by growing recombinant plant rice cells on sterile nutrient medium

w carboh source, aa, mineral etc.

2. under aseptic conditions (in lamina flow hood)

using sterilised apparatus (autoclave @ 121°C for 15 min)

3. place culture in tissue culture room

with opt growth conditions e.g. light intensity / humidity / temp

4. alter PGR conc. using intermediate auxin:CK → callus

↑ auxin:CK → root, ↓ auxin:CK → shoot

[4]

- (ii) Explain why farmers used artificial selection rather than genetic engineering to produce deepwater rice variety with high yield.

1. rice plants are of diff varieties but same species

able to interbreed to produce fertile viable offspring which may inherit the SK1 and SK2 genes

2. artificial selection is cheaper

requires less equipment / no technology

[2]

[Total: 13]

Planning Question

Most mineral elements are essential for the healthy growth of plants. However, lead is known to be toxic and impede plant growth. Lead forms strong covalent bonds with a plant enzyme and its binding alters the 3D conformation of the enzyme's active site.

You are to design an experiment to investigate the effect of different concentrations of heavy metal lead (as lead chloride solution) on the growth of cress seedlings.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you must use:

- 50 cress seedlings
- Petri-dish with cotton wool
- 2.0 mol dm^{-3} lead chloride solution
- distilled water
- glass rod
- pH buffer (pH 6)
- ruler
- lamp
- weighing balance
- a variety of different sized beakers, measuring cylinders, syringes and pipettes for measuring volumes

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

Theoretical background [2m]

1. ref to Pb being a non-competitive inhibitor, binds to allosteric site of enz, binds irreversibly, resulting in 3D config of enz altered, enz involved in metabolic pathways/ PS affected, growth of seedlings reduced;
2. state hypothesis ref to increase in PbCl_2 , less ES cplxes formed, less growth of cress seedlings, lower ave % change in wt of seedlings (after 1 to 7 days)

Independent variable [1m]

3. conc. of PbCl_2 , at least 5 diff conc. with regular intervals b/w each conc. & wide range, vol of 2.0 moldm^{-3} PbCl_2 & vol of distilled water used for dilution shown in dilution table (total vol 10 cm^3)

Dependent variable [1m]

4. average % change in wt of seedlings (after 1 to 7 days) using weighing balance to record fresh wt of seedlings (In g) before & after treatment

Controlled variable (any 2) [1m] rationale must be stated

5. stage of growth of cress seedlings to be the same
6. no. of cress seedlings to be used per petri dish (~ 1 per petri dish)
7. vol. of PbCl_2 to be used (~ $5\text{-}10 \text{ cm}^3$)
8. use of pH buffer (to maintain pH 6)
9. distance of lamp away from petri dish (~ 10 cm)
10. period of time (1-7 days) for soaking of seedlings in PbCl_2

Control experiment [1m] rationale must be stated

11. cress seedlings soaked/watered with water, subject to same experimental conditions

Procedure [2m]

12. coherent sequence of steps e.g. sequence of dilution, weighing initial weight of cress seedlings, soaking in PbCl_2 of diff conc, maintain light intensity of lamp at a fixed distance ~ 10 cm away, illuminate seedlings for 1-7 days, repeat expt
13. conduct expt w/o PbCl_2 , no inhibitor to slow down growth of seedlings, should demonstrate good growth, shows that lead is responsible for slow growth of cress seedlings, measure final fresh wt of all cress seedlings in replicates 7 repeats

Diagram [1m]

14. labelled diagram showing light source & petri dish with PbCl_2 & seedlings

Reliability [1m]

15. 3 replicates & 2 repeats (to be stated in procedure)

Results [1m]

16. table of results with correct headings (ave % change in wt, conc of PbCl_2 / mol dm^{-3}) with repeats & replicates

17. plots theoretical graph (straight line with –ve grad) with correct axes & labels with units (x-axis: conc of PbCl_2 / mol dm^{-3} , y-axis: ave % change in wt)

Safety [1m] (any 1) states hazard, risk, precaution

18. PbCl_2 is toxic, therefore handle with care using gloves

19. throw all cotton wool including seedlings soaked in PbCl_2 in a biohazard bag & dispose with care

20. avoid touching bulb when lamp is in use or when lamp is just switched off

21. handle power plug of lamp & power socket with dry hands to prevent electrocution

ER:

Hypothesis: students related the concentration of lead to impede the growth of the seedling without referring how to measure the growth of the seedling

Results: Majority of the students did not refer to the change in mass/length but took the length and mass of the seedlings after the experiment as the dependent variable.

Reliability: Majority of the students did not perform replicates and repeats in the procedure.

Procedure: Some students gave few hours

5 Free-response question

Write your answers to this question on the separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate,
- must be in continuous prose, where appropriate,
- must be set out in sections (a), (b) etc., as indicated in the question.

(a) Explain the benefits of the human genome project.

[8]

Students need to get at least 3 of the main key points with elaboration

Main Key Pt 1: Molecular medicine (any 3 with elaboration)(max 3m)

1. *earlier detection of genetic predispositions to disease, by identifying mutations that contribute to disease to be used as disease markers;*
2. *provides knowledge on gene seq of disease-causing allele, improving diagnosis of disease, aids in study of prot encoded by these genes;*
3. *improving disease treatment with specific drug design to better target diseased cell, aids in gene therapy & delivery systems for drugs;*
4. *linkage & physical map has made it easier for researchers to locate genes to study their fns;*
5. *detect disease using linkage studies, facilitates study of gene interxn in cplx genetic diseases;*

Main Key Pt 2: Energy sources & environmental applications (any 1 with elaboration) (max 1m)

6. *transfer of knowledge, techniques & capabilities from HGP to microbial studies allows creation of new energy sources e.g. biofuels;*
7. *develop environmental monitoring technology to detect pollutants;*

Main Key Pt 3: Risk assessment (with elaboration) (1m)

8. *assess cancer risks of toxin exposure to individuals based on knowledge of influence of genetics on susceptibility;*

Main Key Pt 4: Bioarchaeology, anthropology, evolution & human migration (any 1 with elaboration) (max 1m)

9. *provides molecular evidence (homology) in study phylogenetic r/s btw humans & o/r org;*
10. *allows study of human migration thru' germline mutations in lineages;*

Main Key Pt 5: DNA forensics (any 1 with elaboration) (max 1m)

11. *identification of individuals in catastrophes, crime;*
12. *paternity testing;*

Main Key Pt 6: Agriculture, livestock breeding & bioprocessing (with elaboration) (max 1m)

13. *transfer of techniques & knowledge help produce disease-, insect-, & drought-resistant crops which are healthier, more productive;*

- (b) With reference to genetic diseases, explain gene therapy treatment, using both viral and non-viral delivery systems. [6]

e.g. viral delivery system: treatment of SCID (max 3m)

1. *where recombinant retrovirus (retroviral vectors) carries functional human ADA gene & integrated within retroviral genome;*
2. *target cells e.g. haematopoietic stem cells (HSCs) are infected by recombinant retrovirus which attach & penetrate → releasing recombinant viral genome into target cell;*
3. *recombinant viral genome is reverse transcribed by RT & integrates/ inserts functional ADA gene into host cell's genome;*
4. *i.e. treated HSCs contain functional gene, can synthesise functional prot e.g. ADA protein;*

e.g. non-viral delivery system: in treatment of cystic fibrosis (CF) (max 3m)

5. *normal, functional CFTR gene inserted into (cationic) liposomes;*
6. *DNA-liposome complex is inhaled using nasal spray to deliver gene to its target cells i.e. lung epithelial cells;*
7. *DNA-liposome complex fuses with cell memb & enters via endocytosis CFTR gene is released into cytoplasm enters nucleus;*
8. *i.e. treated lung epithelial cells contain functional gene, can synthesise functional CFTR prot;*

- (c) Discuss the social implications of genetically modified crop plants. [6]
*Answers should include BOTH positive & negative implications

Positive implications (max 3m)

1. *increased yield in GM crop plants allowing meeting of consumer's food demands/ needs;*
2. *e.g.: insertion of pest-resistance gene (i.e. Bt gene) in crop plant e.g. corn plant → reduce pest infestations & minimization of crop/ harvest losses;*
3. *improved quality in GM crop plants e.g. nutritional value to alleviate/ help improve nutritional deficiencies;*
4. *e.g.: insertion of two β -carotene biosynthesis genes (i.e. psy & crt1 genes) in rice plant → allows synthesis & accumulation of β -carotene (precursor of vit A) in rice endosperm, i.e. alleviating vitamin A deficiency;*

Negative implications (max 3m)

5. *threat to environmental safety e.g. insertion of herbicide-resistant gene (i.e. glyphosate-resistant EPSP gene) in crop plant e.g. cotton plant;*
6. *encourages excessive use of herbicides (e.g. glyphosate) to remove non-herbicide resistant weed plants → selection of 'superweeds' highly invasive*
7. *threat to human safety as vectors used in transforming plant cells contain selectable markers, usually antibiotic resistance genes;*
8. *transformed plant cells when ingested may release intact antibiotic*

resistance gene in gut → transferred to pathogenic bacteria, leading to outbreak;

[Total: 20]